

Geospatial: defined CRS binding, blocked placement

The current shared proposal still needs exact authored definitions before a full placement build can follow it without guessing. This run stops dependent placement and executes the defined CRS discovery and composition behavior in three independent readers.

The editorial revision closes obsolete interface and transform-authoring contradictions. The remaining audit identifies eight narrower missing contracts and one existing 3D-versus-2D example inconsistency. It does not reopen agreed functional outcomes or treat internal implementation methods as standardized data.

Fresh evidence: 15 tests passed. Each reader passed 12 cases and 19 queries, giving 57 reader/query checks. Source layers and fixture bytes remained unchanged. This run executed zero projection or placement jobs and produced no resolved export.

What is already settled

Authored fact or outcome	Existing answer
CRS authority	Complete WKT is authoritative. The typed definition stores uniform token <code>crs:wkt</code> .
Binding and composition	Applied binding uses a CRS reference. Nearest direct binding defines composed subtree scope.
Placement order	Separate CRS position, orientation and scale resolve before ordinary USD transforms.
Source and output	Preserve source facts and adjusted physical placement. The consumer selects the output CRS.
Stage conventions	Writers or assemblers author unit/up-axis correctives. Readers honor ordinary USD data.

Geographic source positions and geographic coordinate queries are supported. Site calibration lives in a derived CRS definition.

Missing authored representation

Audit category	Exact contract still required
G02: placement fields	Names, types, defaults/requiredness, applicability, orientation/scale bases and mutual order.
G05: WKT string normalization	A prescribed serialized normal form that preserves the represented information.
G06: measurement coordinates	An authored association identifying source coordinates, including external assets.
G07: dependency declaration	A carrier and composition rule readable without traversal, including unloaded content.
G09: export sampling	A record distinguishing exported-sample interpolation from original-source resolution.
G10: existing inconsistency	Reconcile the 3D scope statement with two-axis WKT examples without silently broadening scope.

Questions 2, 10, 11, 12 and 14 retain the specific representation work.

Missing observable definitions

Audit category	Agreed outcome and narrower missing meaning
G03: project adjustments	Post-placement adjustments preserve physical placement. Their authored coordinate frame remains undefined.
G04: components and scene domain	E/N/up and geographic queries are settled. Other component mappings and geographic scene frames/bounds remain open.
G08: extent and agreement	State the approximation domain, how the claim is satisfied and how comparable operations are measured.

Operation accuracy, numerical agreement and approximation over an extent are different claims. A full placement-chain accuracy estimate is not a requirement-28 obligation. Internal differentiation, caching or adapter design is implementation freedom. Missing consumer tests are verification work, not new design questions.

A concrete composition result

Queried prim	Nearest direct binding	Returned definition
/World/Data	/World	Geocentric ITRF2020
/World/Independent/Data	/World/Independent	NAD83 grid with NAVD88 height
/World/Unmarked	/World	Geocentric ITRF2020
/Assembly/Independent/Data	/Assembly/Independent	NAD83 grid with NAVD88 height

The independent descendant keeps its own source CRS when the scene is referenced into an assembly.
An unmarked WKT property does not establish a new binding.

What this run executed

Control group	Cases	Queries per reader
References and referenced assembly	2	6
Stronger layer, variant and inherits	3	4
Equivalent composed data	1	2
Unloaded payload	1	2
Missing, broken or invalid definitions	5	5
Total	12	19

Headless OpenUSD, independently written native C++ OpenUSD and a live OV stage each match authored expectations.

Native discovery does not exercise Hydra placement. Live OV discovery does not ingest resolved geometry.

Why earlier passing tests were insufficient

The earlier freeze pinned private experimental answers, then tested code against those answers. A requirements-only intake excluded contradictory design text. Execution results became the headline despite known specification dependencies.

The prior relationship-based placement reader fails a reference-binding fixture from the shared proposal. This run exposes that mismatch and excludes the old placement candidate. Passing tests cannot substitute a private model for the shared data contract.

The default gate now stops dependent execution. It still finishes independently specified checks and delivers their receipt. No previous render, projected coordinate, export or test count is presented as evidence from this run. The remaining blockers are the named proposal contracts, not deferred mechanical work.